Vinayak Tiwari

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Summary

Computer Engineering undergraduate who builds resilient, production-style backend systems — from an LLM gateway with automatic failover and circuit breaking to an autonomous RAG agent orchestrating multiple tools. Comfortable with async Python, LLM APIs, and infrastructure-level reliability patterns, with real output to show rather than courses completed. Looking to build at zero-to-one speed with a team shipping AI-powered products.

Education

Shah & Anchor Kutchhi Engineering College — Mumbai, India
B.Tech in Computer Engineering | CGPA: 8.28/10 — *2023 – Present*

NES Junior College of Arts, Science and Commerce — Mumbai, India
HSC Science | 84.33% — *2021 – 2023*

Certifications

AI-ML Virtual Internship — Remote
AICTE (Ministry of Education) × EduSkills, supported by Google for Developers — *Jun – Aug 2026*

- 8-week structured program covering neural networks with TensorFlow, CNN-based image classification, and object detection with TensorFlow Lite & Model Maker

Technical Skills

AI & Agentic Systems: LangGraph (StateGraph, Tool/Function Calling), Retrieval-Augmented Generation (RAG), Prompt Engineering, Async Streaming (SSE), Circuit Breaker & Failover Design, AI-Assisted Development (Claude, ChatGPT)
LLM APIs & Vector Search: Google Gemini API, ChromaDB, Text Embeddings, Semantic/Cosine Similarity Search
Languages: Python, TypeScript, JavaScript, Java, C, C++, SQL
Backend & APIs: FastAPI, Node.js, Express.js, REST API Design, JWT, Async/Await (httpx, asyncio), Pydantic
Frontend: React 18, TypeScript, Tailwind CSS, React Router, Vite, Axios
Databases: MySQL, MongoDB Atlas, PostgreSQL, Redis (Upstash), H2
Tools & Platforms: Git, GitHub, Docker, Pytest, Postman, Linux, AWS, Render, Railway, Vercel
Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Projects

Cortex — AI Orchestration Gateway
Python, FastAPI, async/await, httpx, Pydantic v2, Redis, Server-Sent Events — Live Demo | GitHub

- Designed and built a resilient async FastAPI gateway unifying two independent LLM providers (Groq, Gemini) behind one normalized interface via an abstract provider pattern (Python `ABC`), so new providers plug in without touching routing code
- Engineered automatic failover routing for both standard and streaming responses — a failed request against the primary provider transparently retries against a secondary with zero caller-visible downtime; streaming failover commits only after a provider's first chunk succeeds, since HTTP headers can't be un-sent mid-stream
- Built a custom circuit breaker from first principles (closed/open/half-open state machine) that trips after repeated provider failures and self-tests recovery after a cooldown, verified through live failure injection
- Added Redis-backed response caching and per-caller rate limiting using atomic `INCR`/`EXPIRE` operations for correctness under concurrent requests, both designed to fail open gracefully if Redis becomes unreachable; deployed live on Render with a connected GitHub pipeline

Nexus — Autonomous Agentic RAG Assistant
Python, FastAPI, LangGraph, Google Gemini, ChromaDB, React 18, TypeScript, Docker — Live Demo | GitHub

- Built an autonomous LangGraph agent that routes user queries across a document-search RAG tool, an order-lookup tool, and a safe AST-based calculator through a single cyclical StateGraph, falling back to a direct response when no tool is needed
- Engineered a grounded RAG pipeline over a persistent ChromaDB store using Google Gemini's `text-embedding-004` embeddings and hierarchical document chunking (PDF/MD/TXT), with application-controlled source citations and a system prompt enforcing hallucination- and prompt-injection-resistant grounding
- Built a full-stack app — FastAPI backend, React 18/TypeScript/Tailwind frontend — containerized with Docker and deployed to Render, backed by 44 passing Pytest unit and integration tests
- Designed dual runtime modes — live Gemini function calling alongside a deterministic offline agent and embedding fallback — keeping the test suite reproducible without external API costs

Relevant Coursework

Data Structures & Algorithms | Artificial Intelligence | Database Management Systems | Object-Oriented Programming | Operating Systems | Computer Networks | Compiler Design